17/12/2025

Code-E

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 NCERT Booster Test Series for NEET-2026_RM(P5)_NBTS-01E **Time : 180 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. The value of $(5.0 \times 10^{-5}) \times (5.0 \times 10^{-8})$ with due regards to significant figures is

(1) 25×10^{-14}

(2) 25.0×10^{-14}

(3) 2.50×10^{-13}

(4) 25.00×10^{-14}

2. Displacement $\left(\vec{r}\right)$ of a particle in x-y plane varies with time t as given by $\vec{r} = \left(t^2\hat{i} + 2\hat{j}\right)$ m. Velocity of particle at t = 1 s is

(1) $\left(2\hat{i} - 2\hat{j}\right)$ m/s

(2) $2\hat{i}$ m/s

(3) $4\hat{i}$ m/s

(4) $\left(4\hat{i} + 2\hat{j}\right)$ m/s

3. Two trains move in opposite direction along parallel roads. Both of them are 100 m long, travelling with a speed of 12.5 m/s. How long will it take for the trains to cross?

(1) 8 s

(2) 4 s

(3) 6 s

(4) 2 s

4. A body of mass m is attached to a vertical spring having spring constant k whose one end is fixed to ceiling. Body is allowed to fall under the gravity when spring is in its natural length. The maximum expansion in the spring will be

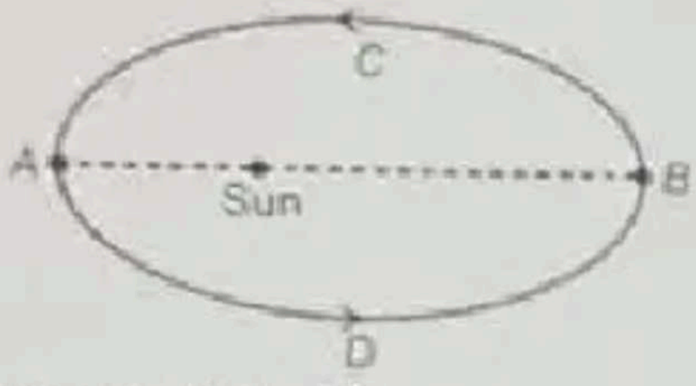
(1) mg/k

(2) $2mg/k$

(3) Zero

(4) $\frac{mg}{2k}$

5. A planet revolves in an elliptical orbit around the sun which is at one of its foci



(I) Gravitational potential energy increases in going from A to B via D.
(II) In path BCA work done by gravity is positive
(III) Angular momentum of planet from B to A via D increases
For the above statements. Choose the correct option.

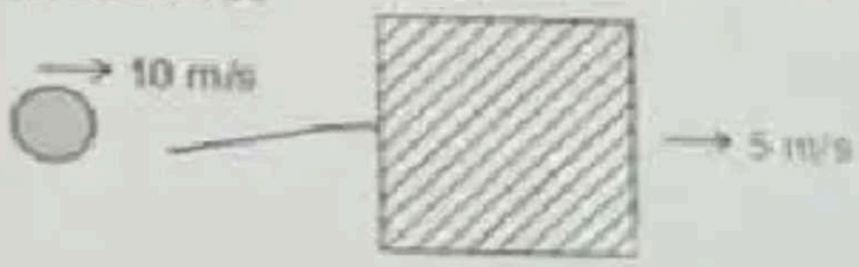
(1) Only statement-(I) and statement-(III) are correct

(2) Only statement-(II) and statement-(III) are correct

(3) Only statement-(I) and statement-(II) are correct

(4) All three statements are correct

6. A light particle moving with a speed 10 m/s collides elastically with a very heavy body moving with a speed 5 m/s as shown in the figure. The speed of the particle after collision will be



*(1) 5 m/s

(2) 10 m/s

(3) 15 m/s

(4) Zero

7. A : Slope of position-time graph gives the velocity
R : Slope of position-time graph and area under acceleration-time graph gives dimensionally same physical quantity.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

8. The weight of an object in a coal mine, at sea level and at the top of mountain are W_1, W_2, and W_3 respectively (depth of coal mine and height of mountain is approximately same). Then

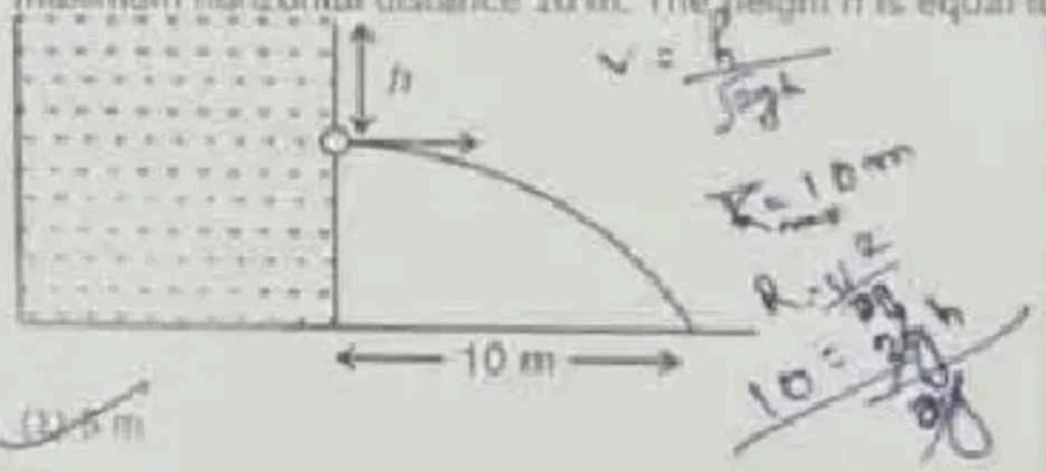
(1) $W_1 > W_2 > W_3$

(2) $W_1 < W_2 < W_3$

(3) $W_1 = W_2 = W_3$

(4) $W_2 > W_1 > W_3$

9. A liquid is coming out from the orifice of tank and falls upto a maximum horizontal distance 10 m. The height h is equal to



(1) 5 m

(2) 8 m

(3) 3 m

(4) 4 m

10. A block of wood floats in water with $\left(\frac{4}{5}\right)^{th}$ of its volume submerged. If the same block floats in a liquid with $\left(\frac{3}{5}\right)^{th}$ of its volume submerged, then density of liquid will be

(1) $\frac{1}{3}$ g cm^{-3}

(2) $\frac{2}{3}$ g cm^{-3}

(3) $\frac{1}{2}$ g cm^{-3}

(4) $\frac{1}{4}$ g cm^{-3}

11. The least count of a screw gauge having 25 divisions on circular scale and 1 mm pitch is

(1) 0.01 mm

(2) 0.02 mm

(3) 0.04 mm

(4) 0.05 mm

12. The equation of trajectory of a projectile in a vertical plane is $y = ax - bx^2$, where a and b are the constants. The horizontal range of the projectile is

(1) a

(2) b

(3) $\frac{a}{b}$

(4) $\frac{b}{a}$

13. A wire can support a load Mg without breaking. It is cut into two equal parts. The maximum load that each part can support is

(1) $\frac{Mg}{2}$

(2) Mg

(3) $\frac{Mg}{4}$

(4) $2Mg$

14. A ring of radius 1 m and mass 5 kg is rotating about its diameter with angular velocity 10 rad/s. Its kinetic energy is

(1) 50 J

(2) 100 J

(3) 125 J

(4) 75 J

15. A : Centre of mass of symmetric bodies with uniform density coincides with their geometrical centre.
R : External force can disturb the motion of centre of mass of system.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

16. A spherical body falling through a viscous liquid of infinite extent finally attains a constant velocity, when

(1) Viscous drag – upthrust = weight

(2) Weight + viscous drag = upthrust

(3) Viscous drag + upthrust = weight

(4) Weight > upthrust + viscous drag

17. If $v = \frac{a}{t} + bt^2$, where v is velocity and t is time. The dimensional formula of b is

(1) $[LT^{-2}]$

(2) $[MLT^{-3}]$

(3) $[LT^{-3}]$

(4) $[LT^{-4}]$

18. Tension T_1 and T_2 in the strings connecting the three blocks as shown in the given figure are respectively



(1) 28 N, 48 N

(2) 48 N, 28 N

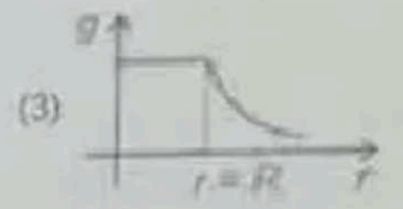
(3) 96 N, 56 N

(4) 56 N, 96 N

19. Assume earth to be a uniform sphere of mass M and radius R. Which of the following graphs best represents the variation of acceleration due to gravity (g) with distance (r) from the centre of the earth?

(1)

(2) 

(3) 

(4) 

20. The SI unit of stress is same as the SI unit of

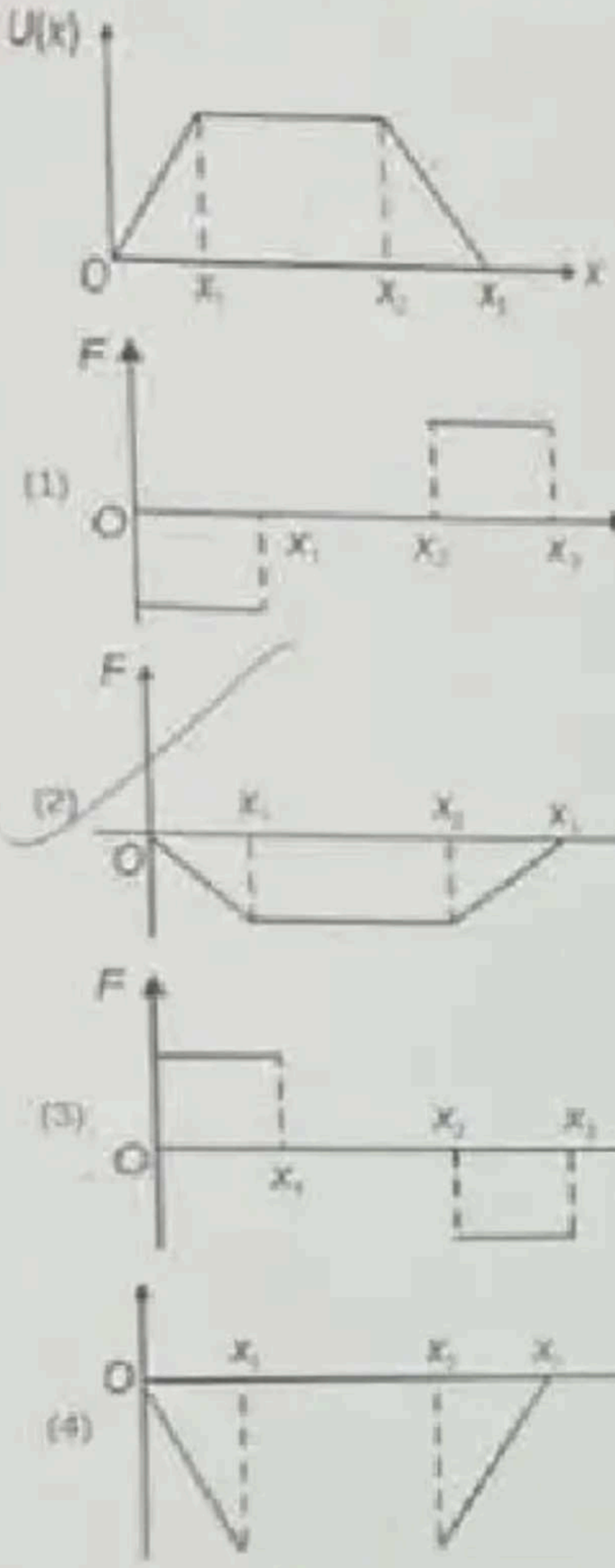
(1) Strain

(2) Modulus of elasticity

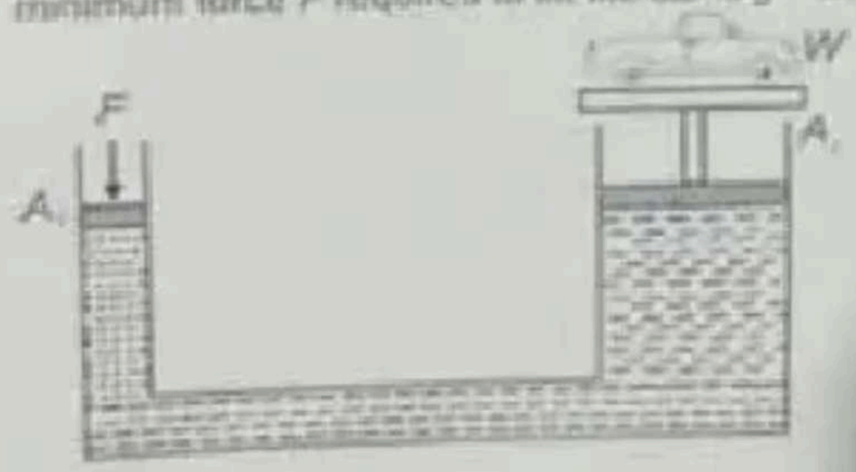
(3) Pressure

(4) Both (2) & (3)

21. The variation of potential energy U of a system is shown in figure. The force acting on the system is best represented by



22. In a hydraulic lift as shown in figure, mass of the car is m = 1000 kg, area of pistons $A_1 = 20\ cm^2$, $A_2 = 5\ m^2$. The minimum force F required to lift the car is $g = 10\ m/s^2$



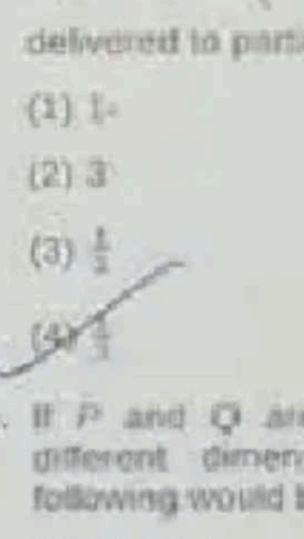
(1) 8 N

(2) 4 N

(3) 6 N

(4) 1 N

23. A man of mass 50 kg stands on a box of mass 30 kg. For the system to be in equilibrium, what force man must exert on the rope?



(1) 400 N

(2) 800 N

(3) 300 N

(4) 1000 N

24. A particle moves with the velocity $\vec{v} = (2\hat{i} - 3\hat{j} + \hat{k})$ m/s under the influence of a constant force $\vec{F} = (2\hat{i} + b\hat{j} - 3\hat{k})$ N. If the instantaneous power delivered to particle is zero, then value of b is

(1) $\frac{1}{3}$

(2) 3

(3) $\frac{1}{2}$

(4) $\frac{2}{3}$

25. If P and Q are two different physical quantities having different dimensional formula, then which among the following would be a valid expression

(1) $P + Q$

(2) $3P - 2Q$

(3) $\frac{P}{Q}$

(4) $P^2 - Q^2$

26. Coefficient of restitution for perfectly inelastic collision will be

(1) $e = 0.5$

(2) $e = 0.3$

(3) $e = 0$

(4) $e = 1$

27. A uniform wire of length l and mass m is suspended from the ceiling and a block of mass m is suspended at the other end of wire. If A is the area of cross section of the wire, then

(1) Tensile stress at any cross section of the wire is $-\frac{2mg}{A}$

(2) Tensile stress at any cross section of the wire is zero

(3) Tensile stress in wire at the point attached to the ceiling is $-\frac{mg}{A}$

(4) Tensile stress at any cross section of the wire is $2\ mg$

28. A system of units has 10 kg, 10 cm and 10 s as the units of mass, length and time respectively. Then the value of 10 N in this system will be

(1) 1000 units

(2) 1 units

(3) 10 units

(4) 100 units

29. A rubber ball is taken to depth 2 km inside water so that its volume reduces by 0.1%, what is the bulk modulus of rubber? ($P_0 = 1.0 \times 10^5$ Pa)

(1) 2×10^{10} N/m^2

(2) 2×10^{11} N/m^2

(3) 2×10^{9} N/m^2

(4) 2×10^{8} N/m^2

30. A particle located at origin at time $t = 0$, starts moving along positive y direction with a velocity v that varies as $v = 2\sqrt{y}$. The displacement of the particle varies with time as

(1) t^3

(2) t^{-2}

(3) t^2

(4) t

31. A man wants to cross the river, in minimum possible path, which is flowing at a speed of 3 km/h. If the man can swim with a speed of 5 km/h in still water, then time taken to cross the river in desired path would be (Width of the river is 500 m)

(1) 1 h

(2) $\frac{1}{3}$ h

(3) $\frac{1}{4}$ h

(4) $\frac{1}{2}$ h

32. The time period of an earth's satellite in circular orbit is independent of

(1) Mass of satellite

(2) Mass of earth and radius of orbit of satellite

(3) Mass of earth

(4) Mass of satellite and radius of orbit of satellite

33. Two small bodies of masses 9m and 16m are placed at a separation R. The gravitational potential on the line joining the two bodies where the gravitational field equals zero, will be (G = Gravitational constant)

(1) $-\frac{7Gm}{R}$

(2) $\frac{Gm}{R}$

(3) $-\frac{8Gm}{R}$

(4) $\frac{6Gm}{R}$

34. A fan accelerating uniformly from rest, rotates through 5 radian during first second. Find the angle rotated during next 2 seconds

(1) 40 rad

(2) 15 rad

(3) 10 rad

(4) 25 rad

35. **Assertion (A)** : Internal forces can change the kinetic energy but not the linear momentum of the system.
Reason (R) : Net internal force on the system is non-zero.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

36. A particle starts from the origin at $t = 0$ with a velocity $\left(10\hat{i}\right)$ m/s and moves in x-y plane with acceleration of $\left(2\hat{i} + 2\hat{j}\right)$ m/s^2. Then at $t = 2$ s, match the entries in column I with column II (in SI unit).

	Column I		Column II
a.	x-coordinate is	(i)	$2\sqrt{3}$
b.	y-coordinate is	(ii)	$2\sqrt{53}$
c.	Speed of the particle is	(iii)	24
d.	Magnitude of acceleration of the particle is	(iv)	4

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iv), b(iii), c(ii), d(i)

37. A body is fired upwards. At half the maximum height, the velocity of the body is 10 m/s. The maximum height attained by body is ($g = 10\ m/s^2$)

(1) 100 m

(2) 10 m

(3) 15 m

(4) 20 m

38. A rectangular film of liquid is extended from (4 cm × 2 cm) to (5 cm × 4 cm). If the work done in increasing area is 3×10^{-4} J, then the surface tension of liquid is

(1) 0.250 N/m

(2) 0.125 N/m

(3) 0.20 N/m

(4) 0.15 N/m

39. A car accelerates from rest at a constant rate 2 m/s^2 for some time and after that it decelerates at a constant rate 3 m/s^2 and come to rest. If total time elapsed is 8 s, then maximum velocity acquired by car will be

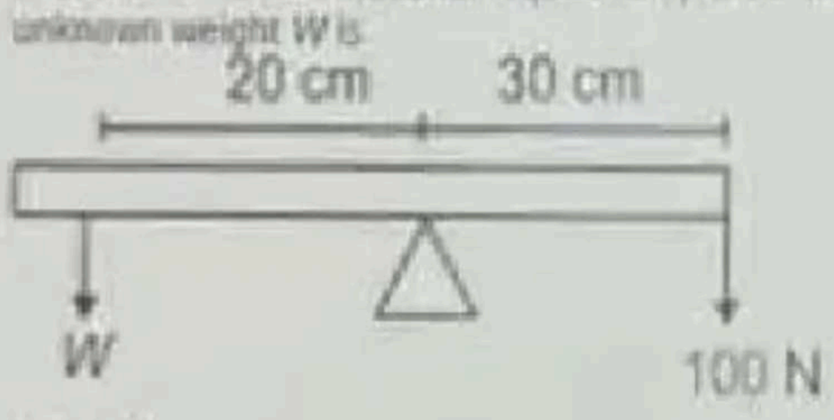
(1) 4.8 m/s

(2) 19.2 m/s

(3) 9.6 m/s

(4) 8.4 m/s

40. If the ruler is in rotational equilibrium, then the value of unknown weight W is



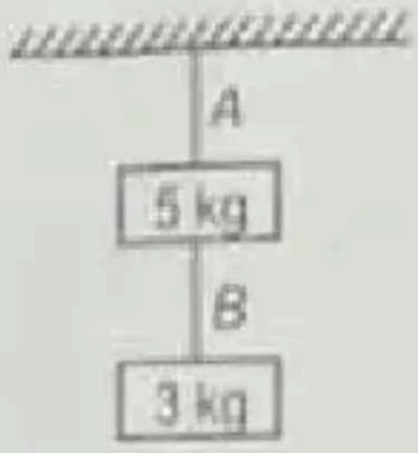
(1) 100 N

(2) 125 N

(3) 90 N

(4) 150 N

41. Two wires A and B of diameter of cross section 0.5 cm each are loaded as shown in figure. The unloaded length of wire A is 3 m and that of wire B is 2 m. If Young's Modulus of both wires is same, then the ratio of elongation of wire A to the wire B will be



(1) 2 : 1

(2) 3 : 2

(3) 4 : 3

(4) 3 : 1

42. The Young's modulus and shear modulus holds relevance for

(1) Solids

(2) Liquids

(3) Gases

(4) Both (1) and (2)

43. When a conservative force does positive work on a body

(1) The potential energy decreases

(2) The potential energy increases

(3) Total energy increases

(4) Total energy decreases

44. The co-ordinates of two points A and B are given respectively as $(0, 4, -2)$ and $(-2, 5, -4)$. The displacement vector from A to B is

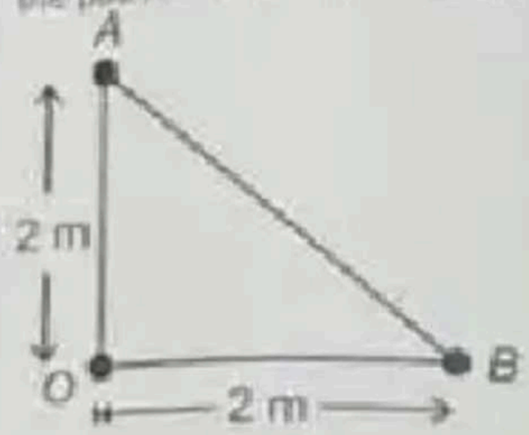
(1) $-2\hat{i}+\hat{j}-2\hat{k}$

(2) $2\hat{i}-\hat{j}+2\hat{k}$

(3) $2\hat{i}+9\hat{j}+9\hat{k}$

(4) $-2\hat{i}-\hat{j}+6\hat{k}$

45. Three identical spheres each of mass m are placed at the corner of right angled triangle with mutually perpendicular side equal to 2 m. Taking their point of intersection as origin, the position vector of centre of mass is



(1) $\frac{1}{3}(i-j)$

(2) $\frac{2}{3}(i-j)$

(3) $\frac{2}{3}(i+j)$

(4) $\frac{1}{3}(i+j)$

CHEMISTRY

46. Given below are two statements: One is labelled as **Assertion A** and other is labelled as **Reason R**.
Assertion A : Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.
Reason R : Energies of the orbitals in the same subshell increases with increase in the atomic number.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

47. Match list-I with list-II and choose the correct option.

List-I (Reactions)	List-II (Relation between ΔH and ΔU)
a. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(i) $\Delta H = \Delta U - \frac{1}{2}RT$
b. $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$	(ii) $\Delta H = \Delta U + RT$
c. $C(s) + O_2(g) \rightarrow CO_2(g)$	(iii) $\Delta H = \Delta U - 2RT$
d. $CO(g) - \frac{1}{2}O_2(g) \rightarrow CO_2(g)$	(iv) $\Delta H = \Delta U$

(1) a(ii), b(iv), c(i), d(iii)

(2) a(i), b(ii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(i), c(iv), d(i)

48. IUPAC official name of element with atomic number 106 is

(1) Bohrium

(2) Seaborgium

(3) Nobelium

(4) Mendelevium

49. The elemental composition of an organic compound was found to be 54.2% C, 9.2% H and 36.6% O. Empirical formula mass of the organic compound is

(1) 66 u

(2) 44 u

(3) 88 u

(4) 22 u

50. Element A has 2 electrons in the outermost orbit and element B has 6 electrons in the outermost orbit. The formula of the compound formed A and B would be

(1) A_2B_3

(2) AB_3

(3) A_2B

(4) A_3B_2

51. If the pH of a saturated solution of sparingly soluble base $B(OH)_2$ is 10 then the value of solubility product (K_{sp}) of $B(OH)_2$ will be

(1) 5×10^{-10}

(2) 5×10^{-13}

(3) 2×10^{-9}

(4) 7.5×10^{-11}

52. 250 mL of 0.1 M aqueous $AgNO_3$ is mixed with 100 mL of 0.2 M NaCl. The amount of AgCl precipitated is (Atomic mass of Ag = 108 u)

(1) 2.12 g

(2) 6.57 g

(3) 2.87 g

(4) 8.23 g

53. If the standard reduction potential of three redox couples A^{2+}/A, B^{2+}/B and C^{2+}/C are -2.1 V, 1.0 V and -0.5 V respectively then the reducing power of A, B and C will be in order of :

(1) B > A > C

(2) C > A > B

(3) B > C > A

(4) A > C > B

54. Identify the conjugate acid and conjugate base in HSO_4^- respectively

(1) SO_4^{2-} and H_2SO_4

(2) H_2SO_4 and SO_4^{2-}

(3) SO_4^{2-} and SO_4^{2-}

(4) H_2SO_4 and SO_4^{2-}

55. Mass of 6 molecules of H_2O is approximately

(1) 18×10^{-23} g

(2) 18×10^{-20} g

(3) 6×10^{-23} g

(4) 12×10^{-20} g

56. The pH of 0.1 M sodium acetate solution is [Given pK_a of acetic acid = 4.76]

(1) 9.8

(2) 8.9

(3) 5.1

(4) 9.4

57. Match batteries in list-1 with their electrolyte in List-2.

	List-1		List-2
(a)	Dry cell	(i)	38% aq H_2SO_4
(b)	Mercury cell	(ii)	Paste of NH_4Cl and $ZnCl_2$
(c)	(H_2+O_2) Fuel cell	(iii)	Paste of KOH and ZnO
(d)	Lead storage battery	(iv)	Conc. (aq) NaOH

(1) (a) (ii), (b) (iii), (c)(i), (d)(iv)

(2) (a) (iii), (b) (ii), (c)(i), (d) (iv)

(3) (a)(iii), (b) (ii), (c) (iv), (d) (i)

(4) (a) (ii), (b) (iii), (c) (iv), (d) (i)

58. Match list-I with list-II.

	List-I		List-II
(a)	Number of orbitals in a subshell	(i)	n
(b)	Number of subshells in a shell	(ii)	$2l + 1$
(c)	Maximum number of electrons in a shell	(iii)	$4l + 2$
(d)	Maximum number of electrons in a subshell	(iv)	$2n^2$

Which of the following options has all correct pairs ?

(1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

59. Correct order of first ionization enthalpy of the given elements is

(1) F > O > N > C

(2) N > F > O > C

(3) O > F > N > C

(4) F > N > O > C

60. The standard electrode potential values for Ag^+/Ag = +0.80 V and Sn^{2+}/Sn = –0.14 V Cr^{3+}/Cr = –0.74 V and Al^{3+}/Al = –1.66 V Identify the correct statement among the following.

(1) Sn^{2+} can oxidise both Al and Cr but not Ag.

(2) Ag is a better reducing agent than Sn.

(3) Cr^{3+} can oxidise Ag but not Al.

(4) Al^{3+} is the strongest oxidising agent among Ag^+, Sn^{2+} and Cr^{3+}.

61. If K_a for an acid is 10^{-6}, then the value of pK_b for its conjugate base is

(1) 6

(2) 8

(3) 10

(4) 14

62. Shape and hybridisation of BrF_5 respectively are

(1) Trigonal bipyramidal and sp^3d

(2) Pentagonal bipyramidal and sp^3d^2

(3) Square pyramidal and sp^3d^2

(4) Trigonal bipyramidal and sp^3d^2

63. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ respectively are

(1) +6 and 3

(2) +6 and 6

(3) +3 and 3

(4) +3 and 6

64. One mole of SO_3 undergoes dissociation according to the given equation. If degree of dissociation of $SO_3(g)$ $\left\{SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)\right\}$ is 20% then value of K_c will be nearly (volume of container = 1 L)

(1) 0.08

(2) 4

(3) 12.5

(4) 25

65. If 14 g of nitrogen gas is mixed with 6 g of hydrogen gas, then the maximum mass of ammonia produced in the reaction is

(1) 17 g

(2) 15 g

(3) 20 g

(4) 12 g

66. The correct order of wavelength of given electromagnetic radiations is

(1) Radio waves > Microwaves > IR > Visible

(2) Microwaves > Radio waves > IR > Visible

(3) Visible > IR > Microwaves > Radio waves

(4) IR > Visible > Microwaves > Radio waves

67. Match reactions in List-I with their type in List-II.

List-I	List-II
(a) $KClO_3 \xrightarrow{\Delta} KCl + O_2$	(i) Non-redox
(b) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	(ii) Non-metal displacement redox
(c) $Na + H_2O \rightarrow NaOH + H_2$	(iii) Metal displacement redox
(d) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$	(iv) Decomposition redox

Correct match is

(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

68. Given below are two statements.

Statement I : The size of anion is larger than that of parent atom.

Statement II : Among isoelectronic species, the one with the larger positive nuclear charge will have a smaller radius.

In the light of above statements choose the correct option given below.

(1) Both statement I and statement II are incorrect

(2) Both statement I and statement II are correct

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

69. Which of the following species is diamagnetic?

(1) O_2^-

(2) O_2^{2-}

(3) O_2

(4) N_2^+

70. The standard enthalpies of formation of $CO_2(g)$, $H_2O(l)$ and glucose(s) at 25°C are –400 kJ/mol, –300 kJ/mol and –1300 kJ/mol, respectively. The standard enthalpy of combustion per gram of glucose at 25°C is

(1) +2900 kJ

(2) –2900 kJ

(3) –16.11 kJ

(4) +16.11 kJ

71. What is the pH of the resulting solution when 100 mL 0.1 M CH_3COOH is mixed with 50 mL of 0.1 M NaOH solution? (pK_a of CH_3COOH = 4.74)

(1) 4.74

(2) 5.04

(3) 4.44

(4) 9.37

72. For the reaction

$Ni(s) + Cu^{2+}(0.2\ M) \rightarrow Ni^{2+}(0.2\ M) + Cu(s)$

$E^\circ_{Cu^{2+}/Cu} = 0.34\ V$, $E^\circ_{Ni^{2+}/Ni} = -0.25\ V$

The EMF of the cell is

(1) 0 V

(2) 0.59 V

(3) 1.18 V

(4) 0.83 V

73. A reaction is spontaneous at all temperatures when

(1) $\Delta_r H < 0$ and $\Delta_r S < 0$

(2) $\Delta_r H < 0$ and $\Delta_r S > 0$

(3) $\Delta_r H > 0$ and $\Delta_r S < 0$

(4) $\Delta_r H > 0$ and $\Delta_r S > 0$

74. If the electrode potential of $A^{2+} + e^- \rightarrow A^+$ is x_1 V and of $A^{2+} + 2e^- \rightarrow A$ is x_2 V then electrode potential of $A^+ + e^- \rightarrow A$ will be

(1) $(x_1 - x_2)$V

(2) $(x_1 + x_2)$V

(3) $(2x_1 - x_2)$V

(4) $(x_2 - x_1)$V

75. For which of the following processes, entropy change is negative?

(1) Evaporation of water

(2) Sublimation of camphor

(3) Condensation of iodine vapours

(4) Melting of ice

76. Suppose the elements X and Y combine to form two compounds X_2Y and X_3Y_4. 0.1 mole of X_2Y weighs 10 g and 0.05 mole of X_3Y_4 weighs 15 g, the atomic masses of X and Y respectively are

(1) 20 u and 60 u

(2) 50 u and 30 u

(3) 60 u and 20 u

(4) 30 u and 50 u

77. According to the Bohr theory, which of the following transitions in the hydrogen atom will give rise to the most energetic photon?

(1) n = 6 to n = 2

(2) n = 5 to n = 1

(3) n = 6 to n = 4

(4) n = 5 to n = 3

78. If conductivity of 10^{-3} M CH_3COOH solution is 1.65×10^{-5} S cm^{-1} and limiting molar conductivity of H^+ and CH_3COO^- ion are 349.6 S cm^2 mol^{-1} and 40.9 S cm^2 mol^{-1} respectively. The degree of dissociation for 10^{-3} M CH_3COOH will be

(1) 4.7%

(2) 2.3%

(3) 11.2%

(4) 9.1%

79. Mass of copper deposited at the cathode when the aq. solution of $CuSO_4$ is electrolysed for 9.65 minute with a current of 10 Ampere is

(1) 30 × 63.5 g

(2) 3 × 63.5 g

(3) 0.03 × 63.5 g

(4) 63.5 g

80. Match the element in column I with that in column II.

	Column I		Column II
(a)	Copper	(i)	Non-metal
(b)	Fluorine	(ii)	Transition Metal
(c)	Silicon	(iii)	Lanthanoid
(d)	Cerium	(iv)	Metalloid

Identify the correct match

(1) (a)-(i) (b)-(ii) (c)-(iii) (d)-(iv)

(2) (a)-(ii) (b)-(iv) (c)-(i) (d)-(iii)

(3) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)

(4) (a)-(iv) (b)-(ii) (c)-(i) (d)-(iii)

81. The number of maximum spectral lines obtained in visible region when an electron jumps from 6th orbit to ground state in hydrogen spectrum is

(1) 1

(2) 2

(3) 3

(4) 4

82. The total number of electrons in 4.2 g of N^{3-} ion is (N_A is the Avogadro's number)

(1) 2.1 N_A

(2) 4.2 N_A

(3) 3 N_A

(4) 3.2 N_A

83. Given below are two statements.

Statement I: Galvanic cell converts chemical energy of a spontaneous redox reaction into electrical energy.

Statement II: Electrode potential is the potential difference developed between the electrode and the electrolyte.

In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

84. Values of n, l, m and s for electron of $4s^1$ orbital will be respectively

(1) 4, 1, 0, $+\frac{1}{2}$

(2) 4, 0, 1, $+\frac{1}{2}$

(3) 4, 0, 0, $+\frac{1}{2}$

(4) 4, 2, 0, $+\frac{1}{2}$

85. Consider the following statements about PCl_5.

(a) The hybridisation and shape of PCl_5 is sp^3d and trigonal bipyramidal respectively.

(b) Three P–Cl bonds lie in equatorial plane and make an angle of 120° with each other.

(c) Equatorial bonds are slightly longer and weaker than axial bonds.

Choose the correct answer from the options given below

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

86. Choose the correct order of increasing bond angles in the following species.

(1) $H_2O < BF_3 < NH_3 < CO_2$

(2) $BF_3 < NH_3 < H_2O < CO_2$

(3) $H_2O < NH_3 < BF_3 < CO_2$

(4) $NH_3 < H_2O < CO_2 < BF_3$

87. If for the reaction, $A(l) \rightarrow B(g)$ at 300 K $\Delta H = 5.7$ kcal mol^{-1}, $\Delta S = 40$ cal K^{-1} mol^{-1}, then ΔG is (R = 2 cal K^{-1} mol^{-1})

(1) −2.5 kcal mol^{-1}

(2) −3.4 kcal mol^{-1}

(3) −6.3 kcal mol^{-1}

(4) −5.1 kcal mol^{-1}

88. Which of the following is a disproportionation reaction?

(1) $2KClO_3 \rightarrow 2KCl + 3O_2$

(2) $CuSO_4 + Zn \rightarrow Cu + ZnSO_4$

(3) $S_8 + 12OH^- \rightarrow 4S^{2-} + 2S_2O_3^{2-} + 6H_2O$

(4) $2F_2 + 2OH^- \rightarrow 2F^- + OF_2 + H_2O$

89. The pair of elements which show diagonal relationship with each other is

(1) Be and Na

(2) Li and Mg

(3) Al and C

(4) Si and N

90. Match the column I containing diatomic molecules with the respective type of bonds present in them.

Column I	Column II
(a) B_2	(i) 2 π bonds only
(b) N_2	(ii) 1 σ and 1 π bond
(c) C_2	(iii) 1 σ and 2 π bonds
(d) O_2	(iv) 1 π bond only

Choose the correct option.

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(ii), (b)-(iii), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

BOTANY

91. The concentration of number of ions and other materials is higher in vacuole than in cytoplasm because
(1) Membrane of vacuole is permeable for all substances
(2) Tonoplast facilitates the transport of ions and other material against the concentration gradient
(3) Vacuole is non-membrane bound structure
(4) Vacuole stores and synthesises ATP to absorb ions and materials

92. State **true (T)** or **false (F)** and choose the **correct** option:
A. Golgi apparatus is the important site of formation of glycoproteins and glycolipids.
B. Reserve materials in the eukaryotic cells are stored in the cytoplasm in the form of inclusion bodies only.
C. Gas vacuoles are absent in blue green algae and purple bacteria but are found in green photosynthetic bacteria.
D. Animal cells have centrioles which are absent in almost all higher plants.

	A	B	C	D
(1)	T	T	F	T
(2)	T	T	T	T
(3)	T	F	T	F
(4)	T	F	F	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

93. Find out the **incorrect** match
(1) SER – Synthesis of lipid and steroid
(2) Polysome – Translate the mRNA into proteins
(3) Centrioles – Basal body which gives rise to cilia and flagella
(4) Nucleolus – Synthesis of tRNA

94. A special type of plastid which store starch is
(1) Chromoplast
(2) Elaioplasts
(3) Aleuroplast
(4) Amyloplast

95. Select the **odd** one w.r.t. function of cell wall in an organism
(1) Maintenance of shape of cells
(2) Protection from mechanical injury
(3) Protection from the attack of pathogens
(4) Involvement in secretion and exocytosis processes

96. A bacterium attaches to the host cell with the help of a bristle-like structures is called
(1) Pili
(2) Flagella
(3) Fimbriae
(4) Cell wall

97. Mesosomes do not help in
(1) Cell wall formation
(2) DNA replication
(3) Photosynthesis
(4) Respiration

98. Read the statements and select the **correct** option.
Statement A: Cytoplasm is the main arena of cellular activities in only plant cells.
Statement B: Ribosomes are non-membrane bound organelles.
(1) Only statement A is correct
(2) Both statements A and B are correct
(3) Only statement B is correct
(4) Both statements A and B are incorrect

99. Synapsis takes place in which of the following stages of meiosis?
(1) Pachytene
(2) Diplotene
(3) Zygotene
(4) Leptotene

100. Duration of cell cycle in yeast is about
(1) 24 hours
(2) 6 hours
(3) 66 minutes
(4) 90 minutes

101. Anaphase I is characterised by
(1) Alignment of chromosomes on equatorial plate
(2) Reappearance of nucleolus and nuclear membrane
(3) Separation of homologous chromosomes
(4) Terminalisation of chiasmata

102. Consider the following events that occur in the M phase of mitosis.
A. Centrosomes starts radiating out microtubules.
B. Spindle fibres attach to kinetochores of chromosomes.
C. Splitting of centromere and separation of chromatids.
D. Complete disintegration of nuclear envelope.
E. Chromosomes are moved to spindle equator and get aligned along metaphase plate.
Arrange the events in the **correct** sequence and choose the option accordingly.

(1) C → A → D → B → E
(2) A → D → B → E → C
(3) A → E → B → C → D
(4) A → D → E → B → C

103. If a cell at G_1 phase has 10 chromosomes and 10 pg of DNA then just after S phase what will be the amount of DNA & number of chromosomes respectively?
(1) 10 & 10
(2) 10 & 20
(3) 20 & 10
(4) 20 & 20

104. A cell in quiescent stage is
(1) Non-proliferating
(2) Metabolically inactive
(3) Actively dividing
(4) A meristematic cell in plants

105. The correctly written scientific name of potato is
(1) *Solanum tuberosum Linn*
(2) *Solanum tuberosum* Linn
(3) *Solanum Tuberosum* Linn
(4) Solanum tuberosum Linn

106. Match the following column-I with column-II and select the **correct** option.

	Column-I		Column-II
(a)	Genus	(i)	Sapindales
(b)	Class	(ii)	Anacardiaceae
(c)	Order	(iii)	Dicotyledonae
(d)	Family	(iv)	*Mangifera*

(1) a(ii), b(iii), c(iv), d(i)
(2) a(i), b(iv), c(iii), d(ii)
(3) a(iv), b(ii), c(i), d(iii)
(4) a(iii), b(ii), c(i), d(iv)

107. Class of fungi in which reproduction is seen only by asexual means is
(1) Ascomycetes
(2) Basidiomycetes
(3) Phycomycetes
(4) Deuteromycetes

108. In five kingdom classification, only single celled eukaryotes are included in
(1) Monera
(2) Protista
(3) Fungi
(4) Plantae

109. Mycoplasma are the organisms that
(1) Have a thick cell-wall
(2) Are the smallest living cell
(3) Cannot survive without oxygen
(4) Are pathogenic only in plants

110. Match the members of Protista given in column-I with their unique features given in column-II and select the appropriate option.

Column-I	Column-II
a. Chrysophytes	(i) Saprophytic protists
b. Dinoflagellates	(ii) Behave like heterotrophs when deprived of sunlight
c. Euglenoids	(iii) Include diatoms and golden algae
d. Slime moulds	(iv) Have cell-wall with stiff cellulose plates on the outer surface

(1) a(iii), b(iv), c(ii), d(i)
(2) a(iii), b(i), c(ii), d(iv)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iii), b(i), c(iv), d(ii)

111. Read the following statements and select the **correct** option.
Statement (A): Archaebacteria differ from other bacteria in having a different cell wall structure.
Statement (B): Chemosynthetic autotrophic bacteria oxidise various inorganic substances and use the released energy for their ATP production.
(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both statements (A) and (B) are correct
(4) Both statements (A) and (B) are incorrect

112. Read the following statements and choose the **correct** option.
A. *Gonyaulax* is a red dinoflagellate.
B. In *Claviceps*, plasmogamy is immediately followed by karyogamy.
C. *Alternaria* asexually reproduces through conidia.

(1) Only A
(2) A and B only
(3) A and C only
(4) Only B

113. State **true (T)** or **false(F)** for the following statements and select the **correct** option.
(A) Dmitri Ivanowsky recognised certain microbes as causal organism of the mosaic disease of tobacco.
(B) W.M. Stanley showed that viruses could be crystallised and crystals consist largely of lipids.
(C) Few members of the kingdom plantae are partially heterotrophic.
(D) *Contagium vivum fluidum* was coined by Beijerinck.

	A	B	C	D
(1)	F	T	T	F
(2)	T	T	F	T
(3)	T	F	T	T
(4)	F	F	T	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

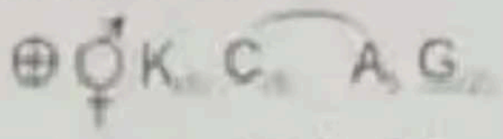
114. Match the **Column I** with **Column II** and select the **correct** option.

Column I	Column II
a. Basal placentation	(i) Primrose
b. Axile placentation	(ii) Mustard
c. Parietal placentation	(iii) Tomato
d. Free central placentation	(iv) Marigold

(1) a(iv), b(ii), c(iii), d(i)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(iv), b(iii), c(i), d(ii)
(4) a(iv), b(iii), c(ii), d(i)

115. Coconut fruit differs from mango fruit as the former
(1) Is a drupe fruit
(2) Has edible endosperm
(3) Has stony endocarp
(4) Develops from monocarpellary ovary

116. The floral formula



belongs to which family?
(1) Brassicaceae
(2) Solanaceae
(3) Asteraceae
(4) Fabaceae

117. The example of hypogynous flower is
(1) Petunia
(2) Rose
(3) Guava
(4) Peach

118. In maize seed, single large and shield shaped cotyledon known as
(1) Coleoptile
(2) Coleorhiza
(3) Scutellum
(4) Hilum

119. How many among the following flowers are actinomorphic?
Pea, Mustard, Gulmohur, *Cassia*, Datura, Bean, Chilli
(1) Four
(2) Three
(3) Six
(4) Two

120. Which of the following shows whorled phyllotaxy?
(1) *Alstonia*
(2) *Calotropis*
(3) Mustard
(4) China rose

121. Given below are two statements : One is labelled as Assertion **(A)** and the other is labelled as Reason **(R)** :
Assertion (A) : A flower is defined as a modified shoot, wherein the shoot apical meristem changes to floral meristem.
Reason (R) : Internode of the shoot gets condensed to produce different floral appendages, laterally at successive nodes, instead of leaves.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true but R is not the correct explanation of A
(2) A is true but R is false
(3) A is false but R is true
(4) Both A and R are true and R is the correct explanation of A

122. Select the **correct** statement regarding monocot stem.

(1) Hypodermis is sclerenchymatous

(2) Endodermis is rich in starch

(3) Pith is parenchymatous with spaces

(4) Vascular bundles are conjoint and open

123. State **true (T)** or **false (F)** to the following statements and select the correct option.

A. In dicot leaves, the vascular bundles can be seen in the veins and the midrib.

B. In dicot leaves, the size of the vascular bundles is dependent on the size of the veins.

C. The adaxial epidermis bears more stomata than the abaxial epidermis in dicot leaves.

	A	B	C
(1)	T	F	T
(2)	F	F	T
(3)	T	T	F
(4)	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

124. Cortex is the region found between

(1) Epidermis and stele

(2) Pericycle and endodermis

(3) Endodermis and pith

(4) Endodermis and vascular bundle

125. Cells of the outer-most covering of the whole plant body

(1) Are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole

(2) Have a deposition of water-impermeable waxy material lignin

(3) Are not interconnected with each other due to lack of plasmodesmatal connections

(4) Are sclerenchymatous with hard cell walls as they protect the plants from mechanical injuries

126. Read the given statements (A-D) and choose the option with **incorrect** one(s).

A. Roots of grasses differ from roots of sunflower, as the former lack suberin depositions in the innermost layer of cortex.

B. In a typical young dicot stem, the outermost protective layer may bear trichomes and a few stomata.

C. Ground tissue system forms the main bulk of the plant body.

D. Monocotyledonous and dicotyledonous plants show marked variations in their internal structures but they do not differ in type and location of vascular bundles.

(1) B and C only

(2) All A, B, C and D

(3) A and D only

(4) D only

127. Where do you expect to find pericycle in the form of semi-lunar patches made up of sclerenchymatous cells?

(1) Dicot stem

(2) Monocot stem

(3) Dicot root

(4) Monocot root

128. The epidermis in a dorsiventral leaf

(1) Is not covered by cuticle

(2) Covers both adaxial and abaxial surfaces

(3) Is also known as epiblema

(4) Lacks living cells

129. Haploid spores in bryophytes germinate to produce

(1) Free-living gametophyte

(2) Diploid gametophyte

(3) Free-living sporophyte

(4) Sporophyte attached to the parent plant

130. Floridean starch is stored food in which of the following algae?

(1) *Polysiphonia*

(2) *Ectocarpus*

(3) *Laminaria*

(4) *Sprogyra*

131. Select the **correctly** matched pair.

(1)	Cycas	–	Male and female cones are borne on different trees
(2)	*Selaginella*	–	Microphyllous leaves and heterosporous
(3)	*Funaria*	–	A liverwort, used as fuel and as a packing material
(4)	*Ulothrix*	–	Colonial alga and isogamous

(1) (1)

(2) (2)

(3) (3)

(4) (4)

132. The system of classification given by Linnaeus was

(1) Artificial system

(2) Natural system

(3) Phylogenetic system

(4) Based on economic uses of plants

133. Read the following statements (a-d) and answer the question which follows them.

(a) The sporophyte in mosses is more elaborate than in liverworts.

(b) *Volvox*, *Chlamydomonas* and *Euglena* show oogamous type of reproduction.

(c) Free-living gametophytes are found in gymnosperms.

(d) Cycas has branched stem and needle-like leaves

How many of the above statements is/are **correct**?

(1) Two

(2) Three

(3) Four

(4) One

134. Which of the following features is **true** regarding the first terrestrial plants possessing the vascular tissues?

(1) Dominant gametophytic phase

(2) Non-motile male gametes

(3) Presence of compact sporophylls in all the members

(4) Occurrence of photosynthetic thalloid gametophyte as the life cycle called prothallus

135. Read the following statements and choose the **correct** option.

(A) The members of chlorophyceae are usually grass green in colour due to dominance of pigments chlorophyll 'a' and 'b'.

(B) In red algae, complex post-fertilisation developments occur

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both statements (A) and (B) are correct

(4) Both statements (A) and (B) are incorrect

ZOOLOGY

136. Which of the following is **not** a characteristic feature of biceps muscle?

(1) We are usually able to make a contract merely by thinking about it

(2) Its muscle fibres taper at both the ends.

(3) Its muscle fibres are bundled together in parallel fashion.

(4) It has alternate light and dark bands giving a characteristic striped or striated appearance.

137. Read the following statements A and B and choose the **correct** option.

Statement A : Areolar tissue is the most widely distributed loose connective tissue in the animal body.

Statement B : Tendons are elastic structures and are a type of specialized connective tissue.

(1) Statement A is correct but statement B is incorrect

(2) Statement A is incorrect but statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

8. _____ facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.
Choose the option which fills the blank **correctly**.

(1) Tight junctions

(2) Gap junctions

(3) Adhering junctions

(4) Desmosomes

9. Read the following statements.
Statement A : Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres.
Statement B : Most of the cartilages in vertebrate embryos are replaced by bones in adults.
Choose **the correct** option

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

10. Select the **correct** match.

(1)	Simple squamous epithelium	Bowman's capsule
(2)	Simple cuboidal epithelium	Gastric mucosa
(3)	Simple columnar epithelium	PCT
(4)	Simple ciliated epithelium	Intestinal mucosa

(1) (1)

(2) (2)

(3) (3)

(4) (4)

11. Match column I with column II.

Column I		Column II
a.	Acidic amino acid	(i) Valine
b.	Basic amino acid	(ii) Glutamic acid
c.	Neutral amino acid	(iii) Lysine
d.	Aromatic amino acid	(iv) Phenylalanine

Select the **correct** option.

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iv), b-(ii), c-(ii), d-(iii)

(3) a-(ii), b-(iii), c-(i), d-(iv)

(4) a-(iv), b-(iii), c-(ii), d-(i)

142. Fats have A melting point and remain as B at room temperature while oils have C melting point and remain as D at room temperature.

	A	B	C	D
(1)	High	Liquid	Low	Solid
(2)	Low	Liquid	High	Solid
(3)	Low	Solid	High	Liquid
(4)	High	Solid	Low	Liquid

(1) (1)

(2) (2)

(3) (3)

(4) (4)

143. With the increase in substrate concentration, the velocity of the enzymatic reaction rises and ultimately reaches a maximum velocity. What happens to the enzymatic activity when more substrate is added in this condition?

(1) It will decrease

(2) It will increase

(3) It will remain unchanged

(4) It will first increase then it will decrease

144. Choose the **incorrect** match.

(1)	Catalase	–	Haem is a prosthetic group and is a part of active site
(2)	Carboxypeptidase	–	Zinc is a cofactor which forms coordination bonds
(3)	Nicotinamide adenine dinucleotide	–	Contains vitamin niacin
(4)	Peroxidase	–	Contains hydrogen peroxide as prosthetic group

(1) (1)

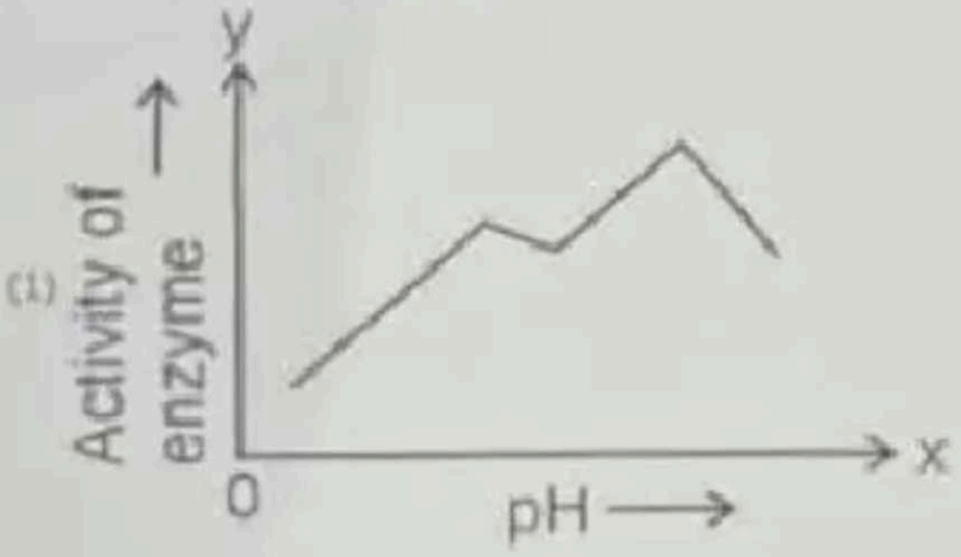
(2) (2)

(3) (3)

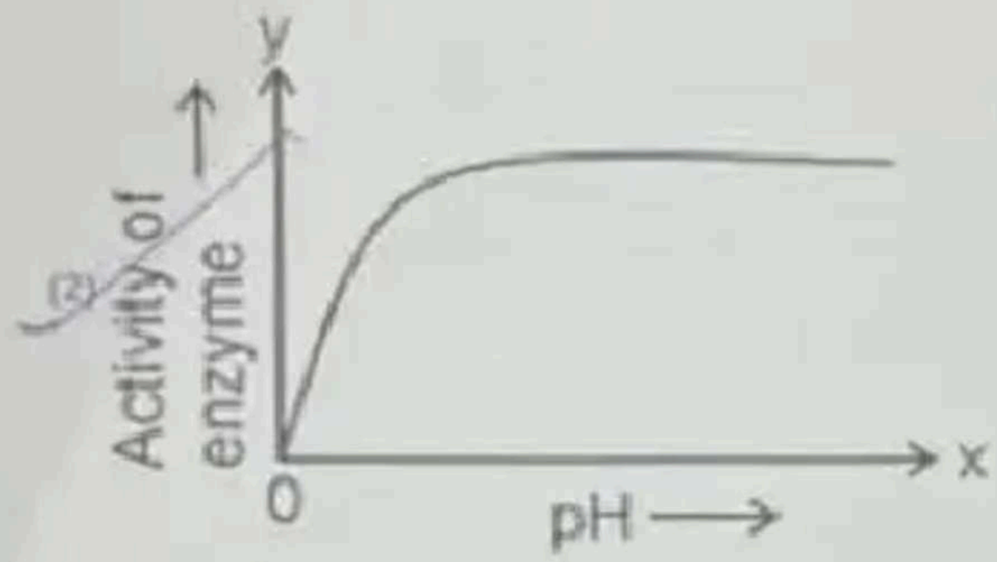
(4) (4)

145. Following graphs represent activity of enzyme versus pH. Choose the graph which **correctly** represents the effect of pH on the activity of enzyme.

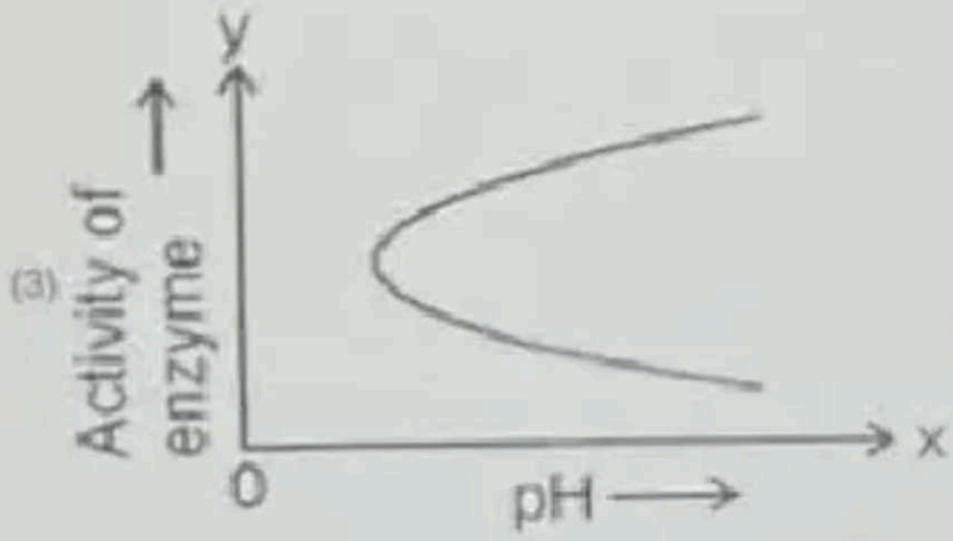
(1)



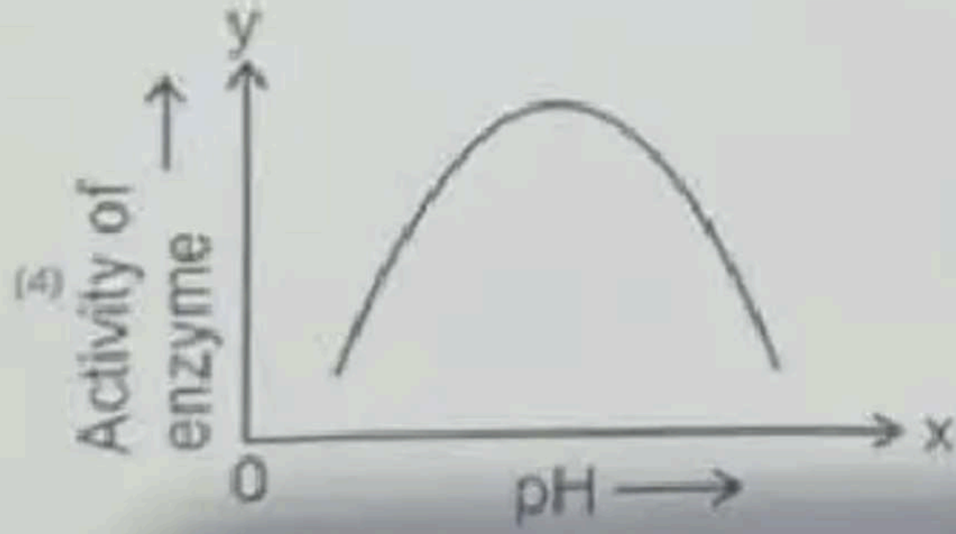
(2)



(3)



(4)



146. The major component which is present in plant cell wall does not give colour with iodine because it notdoes

(1) A homopolymer

(2) A polysaccharide

(3) Not contain complex helices

(4) Not have reducing end

147. Match the following Columns I and II w.r.t partial pressure (mmHg) of O_2.

	Column I		Column II
a.	Atmospheric air	(i)	40
b.	Blood in systemic arteries	(ii)	104
c.	Alveoli	(iii)	159
d.	Tissue	(iv)	95

Select the correct option.

(1) a(iii), b(iv), c(ii), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

148. Read the following
(a) Highly vascular
(b) Lined by squamous epithelium
(c) Irregular walls
The above points **correctly** describe

(1) Terminal bronchioles

(2) Alveoli

(3) Trachea

(4) Primary bronchi

149. The factors which favour the formation of oxyhaemoglobin and leads to the left shift of the oxygen dissociation curve are

(1) High pO_2, high temperature, high H^+ concentration

(2) High pO_2, low temperature, high pH

(3) High pO_2, high pCO_2, low pH

(4) High pO_2, low pH, low temperature

150. The structure among the following which forms the part of the conducting zone of human respiratory system and is not supported by incomplete cartilaginous rings is

(1) Initial bronchiole

(2) Trachea

(3) Tertiary bronchus

(4) Terminal bronchiole

151. Body of a person contains 5 L blood. What amount of O_2 can be delivered by 5 L of blood to tissues under normal physiological conditions?

(1) 750 mL

(2) 500 mL

(3) 250 mL

(4) 150 mL

$\frac{5 \times 5000}{100}$

$= 250 ml$

...humans, the contraction of which of the following muscles causes an increase in the volume of thoracic chamber in the anteroposterior axis during normal breathing?

(1) Abdominal muscles

(2) External inter-costal muscles

(3) Internal inter-costal muscles

(4) Diaphragm

...Match the column-I with column-II. Select the **correct** option.

	Column-I		Column-II
a.	Total lung capacity	(i)	TV + IRV
b.	Vital capacity	(ii)	VC + RV
c.	Functional residual capacity	(iii)	ERV + TV + IRV
d.	Inspiratory capacity	(iv)	ERV + RV

(1) a(iii), b(ii), c(iv), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iv), b(i), c(ii), d(iii)

154. Select the parameter that does **not** undergo change upon strenuous exercise in a person.

(1) Duration of cardiac cycle

(2) Stroke volume

(3) Cardiac output

(4) Site where cardiac impulse originates

155. Read the following statements carefully

(a) Interstitial fluid has the same mineral distribution as that of plasma.

(b) Tissue fluid helps in exchange of gases.

(c) Fats are absorbed through blood in lacteals.

Choose the correct option

(1) Only statement (c) is correct

(2) Only statements (a) and (b) are correct

(3) All statements (a), (b) and (c) are correct

(4) Only statements (b) and (c) are correct

156. What would be the cardiac output of a person if heart rate is 50 beats per minute, blood volume in the ventricles at the end of diastole is 120 mL and at the end of ventricular systole is 50 mL?

(1) 3 L per minute

(2) 5 L per minute

(3) 1.5 L per minute

(4) 6 L per minute

157. Choose the **incorrect** match.

(1)	Cardiac arrest	–	Heart stops beating
(2)	Angina pectoris	–	Chest pain due to myocardial ischemia
(3)	Atherosclerosis	–	Narrowing of coronary artery
(4)	Heart failure	–	Heart muscle is suddenly damaged by inadequate blood supply

(1) (1)

(2) (2)

(3) (3)

(4) (4)

158. In a standard ECG, the contraction of ventricles starts shortly after 'A' and end of 'B' wave marks the end of ventricular systole.
Choose the option that correctly identifies 'A' and 'B' respectively.

(1) P, Q

(2) R, S

(3) Q, T

(4) S, T

159. **Assertion (A):** In humans, the mature RBCs are enucleated and biconcave disc shaped.
Reason (R): Absence of nucleus provides more space for oxygen binding pigment haemoglobin and biconcave shape increases the total surface area for exchange of gases.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

160. Pulmonary vein and pulmonary artery in humans differ from each other in all of the following, **except** .

(1) Thickness of their respective tunica media

(2) Partial pressure of oxygen in the blood that they carry

(3) Number of layers in their walls

(4) Diameter of their lumen

161. For hemodialysis of uremic patients, all of the following are present in the dialysing fluid, **except**

(1) Nitrogenous wastes

(2) Glucose

(3) Electrolytes

(4) Water

162. In humans, under normal physiological conditions, the fluid which is present throughout the distal convoluted tubule has the osmolarity

(1) Lower than the blood plasma

(2) Same as that of the blood plasma

(3) Higher than the blood plasma

(4) Same as that of the innermost medullary interstitium

163. **Assertion (A):** ANF works in response to high blood volume and high blood pressure.
Reason (R): ANF acts as a check on the renin-angiotensin mechanism and decreases blood volume and blood pressure.
In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

164. All of the following organisms excrete their nitrogenous wastes in the form of pellet or paste with a minimum loss of water, **except**

(1) *Calotes*

(2) *Columba*

(3) *Corvus*

(4) *Canis*

165. Select the **incorrect** match among the following w.r.t the kidney of a healthy adult human.

(1) Length – 10 - 12 cm

(2) Width – 5 - 7 cm

(3) Thickness – 2 - 3 cm

(4) Position – Between the levels of last thoracic and first lumbar vertebrae

166. In a normal healthy adult human, the amount of blood filtered by the kidneys per minute is 'X' of the blood pumped out by each ventricle of the heart in a minute. Here, 'X' is

(1) One-fifth

(2) One-third

(3) Two-third

(4) Half

167. Read the following statements:
(a) Myasthenia gravis is an autoimmune disorder characterised by rapid spasms in muscles.
(b) Gout is a type of arthritis caused due to accumulation of uric acid crystals in joints.
(c) During menopause, females are more susceptible to osteoporosis as compared to fertile females.
Choose the option with only correct statements.

(1) a, b and c

(2) Only a and b

(3) Only b and c

(4) Only c

168. Choose the **incorrect** option w.r.t. the total number of different bones that constitute the adult human skeleton.

(1) Cervical vertebrae – 7

(2) Scapula – 2

(3) Ribs – 24

(4) Pelvic girdle – 4

169. Flagellar movements help in all of the following activities, **except**

(1) Maintenance of water current in the canal system of *Sycon*

(2) Swimming of human spermatozoa

(3) Locomotion of *Euglena*

(4) Movement of food through cytopharynx in *Paramoecium*

170. How many structures from the box given below is/are not included in axial skeleton distributed along the main axis of the body?

Skull, Vertebral column, Ribs, Sternum, Femur

(1) Four

(2) Three

(3) Zero

(4) One

171. Each organised skeletal muscle in our body is made of a number of muscle bundles held together by a common collagenous connective tissue layer, called

(1) Fascicle

(2) Fascia

(3) Muscle fibre

(4) Myofibril

a sarcomere, X bisects 'I' band whereas thick laments in the 'A' band are held together in the middle by a ion fibrous membrane called Y.
dentify X and Y.

	X	Y
(a)	'M' line	'Z' line
(b)	'Z' line	'M' line
(c)	'H' zone	'Z' line
(d)	'M' line	'H' zone

Select the **correct** option.

(1) (a)

(2) (b)

(3) (c)

(4) (d)

On the basis of following characters, identify 'A' and 'B' in a human.
A. Pliable matrix due to chondroitin salts
B. Hard matrix due to calcium salts
Select the correct option.

(1) A-Cartilage present at the joint between 8th and 7th rib; B-Sternum

(2) A-Fibula, B-Sternum

(3) A-Hyaline cartilage, B-Fibrous cartilage

(4) A-Fibula, B-Pubic symphysis

4. How many of the following features is/are true w.r.t. electrical synapse?
(a) Transmission of impulse across this is very similar to impulse conduction along a single axon.
(b) It is rare in human system
(c) Impulse transmission across this is always faster than chemical synapse
(d) Impulse can travel directly from one neuron to another
Select the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

5. **Assertion (A)**: During the resting state, axoplasm contains high concentration of K^+ and low concentration of Na^+ as compared to extracellular fluids.
Reason (R): Ionic gradient across the resting axonal membrane is maintained by the active transport of K^+, Na^+ and large negatively charged proteins.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

176. Read the following statements.
(A) Multipolar neurons are found in brain and spinal cord.
(B) Unipolar neurons are found in dorsal root ganglion of the spinal cord.
(C) Pseudo-unipolar neurons are found only in early embryonic stages.
(D) Bipolar neurons have one axon and two dendrites.
Choose the option that states them as true (T) or false (F) correctly.

	(A)	(B)	(C)	(D)
(1)	T	F	T	F
(2)	F	F	F	T
(3)	T	T	F	F
(4)	T	F	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

177. Nerve fibres without nodes of Ranvier can be found in
(a) Autonomous neural system
(b) Somatic neural system
(c) Peripheral neural system
Choose the **correct** option.

(1) a and b only

(2) a, b and c

(3) b and c only

(4) a and c only

178. Body temperature is regulated by which part of human brain?

(1) Hypothalamus

(2) Medulla oblongata

(3) Pons

(4) Cerebrum

179. Consider the following statements about human nervous system.

a. The CNS lies along the central axis of the body

b. CNS includes the brain and the spinal cord

c. The nerves of the body associated with CNS are the components of the PNS.

d. CNS sends regulatory impulses to smooth muscles and glands.

Choose the option that contains only **correct** statements.

(1) a and b only

(2) a and d only

(3) a, b and c only

(4) a, b, c and d

180. Choose the **correct** statement about unmyelinated nerve fibres.

(1) They have a myelin sheath around their axon.

(2) Nodes of Ranvier are present.

(3) Their axon is enclosed by Schwann cells in PNS.

(4) Nerve impulse conduction is faster than in myelinated nerve fibres.